STANDARD OPERATING PROCEDURE (SOP)

Wa CRM — External Proforma & Client API

Developer Implementation Manual & Technical Integration Reference

Document ID:	SOP-INT-WACRM-001	**Effective Date:**	October 2026
Production URL:	`https://wacrm.dinuping.com`	**Version:**	1.0.0 (Production)
API Endpoint:	`POST /api/integrations/proforma`	**Security:**	Pre-shared API Key (Header)

1. Purpose & Authentication Credentials

This API enables external websites, billing systems, and ERPs to securely transmit client profiles and Proforma Invoices into Wa CRM. All incoming requests must be authenticated using the following credentials:

Parameter	Value	Notes
Base URL	`https://wacrm.dinuping.com`	Production endpoint (HTTPS enforced)
Primary Header	`X-API-Key`	Recommended authentication header
API Key Value	`wacrm_ext_live_8849204128f73a0e691823c91`	Must be kept secure in backend .env
Alternative Header	`Authorization: Bearer <KEY>`	Standard OAuth Bearer format
Rate Limit	**120 req / minute**	Sliding window per IP/key (HTTP 429 on breach)

2. API Routes Specification

`POST`	`/api/integrations/proforma`	**Main Endpoint**: Ingests client & proforma, recalculates GST, enforces idempotency, returns authoritative totals.
`GET`	`/api/integrations/proforma/{id_or_num}`	Retrieves single authoritative proforma by numeric ID or Document Number (e.g. PI-2026-0042).
`GET`	`/api/integrations/proforma`	Lists and paginates authoritative proformas with search and customer ID filters.
`OPTIONS`	`/api/integrations/proforma`	CORS preflight handler for web browser and client applications.

3. Duplicate Prevention & Authoritative Calculation

Idempotency Key: Every request should provide a unique UUIDv4 either via header X-Idempotency-Key or body field idempotencyKey. If a request with the same key or proforma number is retried within 24 hours, the CRM returns the previously issued document (processingStatus: AUTHORITATIVE_VERIFIED) without creating duplicate records.

Statutory Tax Rules: The CRM calculates: Taxable Subtotal = (Rate * Qty) - Discount.
• **Intra-state (Tamil Nadu, State Code '33')**: 9% CGST + 9% SGST.
• **Inter-state (Other States)**: 18% IGST.

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If arbitrary client totals disagree with CRM re-computations, the authoritative CRM total is enforced and an advisory warning is returned.

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4. JSON Request Field Dictionary

Field	Type	Req?	Description
`idempotencyKey`	string	Yes*	Unique transaction UUID (prevents duplicate documents).
`companyName`	string	Yes**	Legal company or customer trade name.
`contactPerson`	string	Yes**	Primary contact / signatory name (**Either company or contact required).
`phone`	string	Yes	10-digit phone number with optional country code.
`email`	string	Opt	Email address for billing and invoice dispatch.
`gstin`	string	Opt	15-character GSTIN (e.g. 33AAAAA1234A1Z5).
`billingAddress`	object	Opt	Object with line1, city, state, stateCode, pincode.
`proformaNumber`	string	Opt	Document number. If omitted, CRM auto-generates PI-YYYY-XXXX.
`productCategory`	string	Opt	wahrms (Software) or biometric (Hardware).
`lineItems`	array	Yes	List of purchased items (minimum 1 item required).
`item.name`	string	Yes	Product / Service bundle title.
`item.hsn`	string	Opt	HSN/SAC code (998314 for software, 85437099 for biometric).
`item.qty`	number	Yes	Quantity (must be > 0).
`item.rate`	number	Yes	Unit rate excluding tax (must be ≥ 0).
`item.gstRate`	number	Opt	Tax decimal (0.18) or percentage (18).

5. Live Request & Response Examples

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6. Authoritative Success Response (HTTP 201 Created)


```
HTTP/1.1 201 Created Content-Type: application/json { "success": true, "requestId": "req_88a91b2c4d", "processingStatus": "CREATED", "customerId": 101, "customerName": "Rajesh Kumar", "companyName": "Acme Global Solutions Pvt Ltd", "proformaId": 204, "proformaNumber": "PI-2026-0042", "authoritativeSummary": { "taxableSubtotal": 75000.0, "taxMode": "intra", "cgst": 6750.0, "sgst": 6750.0, "igst": 0.0, "totalTax": 13500.0, "roundOff": 0.0, "grandTotal": 88500.0, "lineItemsCount": 1 }, "pdfDownloadUrl": "/api/integrations/waquote/proformas/204/download", "syncedAt": "2026-10-17T11:10:00+00:00", "message": "Proforma Invoice #PI-2026-0042 successfully registered in CRM." }
```


7. Code Implementation Snippets

Node.js / Axios:

```
const axios = require('axios'); const response = await axios.post('https://wacrm.dinuping.com/api/integrations/proforma', { idempotencyKey: `idem_${Date.now()}`, companyName: 'Acme Corp', contactPerson: 'Ravi', phone: '+91 9876543210', lineItems: [{ name: 'WAHRMS Software', qty: 1, rate: 50000, gstRate: 0.18 }] }, { headers: { 'X-API-Key': 'wacrm_ext_live_8849204128f73a0e691823c91', 'Content-Type': 'application/json' } }); console.log('Invoice Created:', response.data.proformaNumber, 'Total:', response.data.authoritativeSummary.grandTotal);
```

cURL (Terminal Test Command):

```
curl -X POST "https://wacrm.dinuping.com/api/integrations/proforma" \ -H "X-API-Key: wacrm_ext_live_8849204128f73a0e691823c91" \ -H "Content-Type: application/json" \ -d '{"companyName":"Test Client","contactPerson":"Ravi","phone":"9876543210","lineItems":[{"name":"Item","rate":50000,"qty":1,"gstRate":0.18}]}'
```

8. Troubleshooting & HTTP Error Reference

401	Missing server-to-server authentication...	Add header X-API-Key: wacrm_ext_live_8849204128f73a0e691823c91
403	Invalid API key or unauthorized credential	Verify key characters. Check for trailing spaces or typo.
422	Validation failed (missing company/items)	Ensure companyName or contactPerson and lineItems are non-empty.
429	Rate limit exceeded (120 req/min)	Reduce call frequency or implement exponential backoff.

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